

Quantitative Research Report

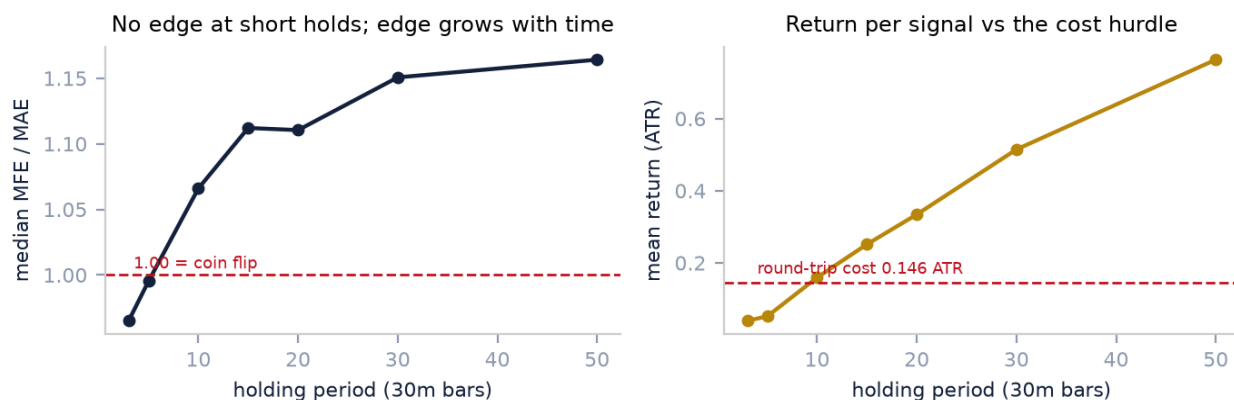
Gold Trend — Strategy | XAUUSD 30m | 78,695 bars, 1 Dec 2019 – 30 Jul 2026 | 4,036 raw signals, 737 traded positions

VERDICT

Moderate evidence of a robust edge - with one disqualifying fragility

1. What the entry signal is actually worth

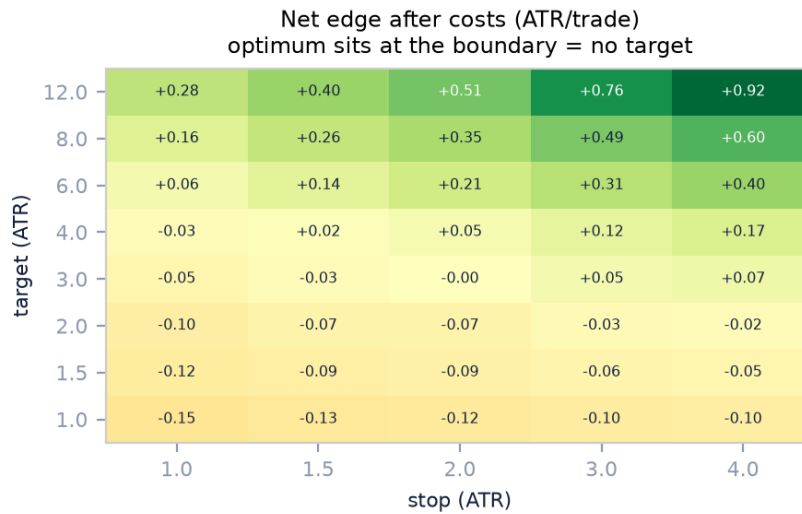
Signal confirms on the close of bar N; entry is the OPEN of bar N+1. No intrabar path of the signal bar is used and no measurement window reads beyond its own horizon. Longs and shorts are kept separate throughout.



The signal has no short-horizon edge. MFE/MAE is 0.97 at a three-bar hold and the signal moves favourably before adversely only **48.8%** of the time — a coin flip. The ratio climbs to 1.16 by 50 bars and mean return climbs from 0.039 to 0.764 ATR. **The edge accrues to hold time, not to entry timing.** That is the same exponent gap this project measured indirectly, now measured on the entry signal itself.

2. Target and stop testing — 103 combinations

Each signal's forward path is walked once and thresholds resolved from it. When a stop and target are first crossed on the SAME 30-minute bar the true order is unknowable without lower-timeframe data; those resolve as LOSSES (conservative) and the frequency is reported.



Ambiguity frequency and why it does not change the conclusion

Setting	Same-bar ambiguity	Effect on the verdict
target 0.25 / stop 0.25	44.3%	severe - but this cell loses heavily anyway
target 1.00 / stop 1.00	1.7%	negligible
target 3.00 / stop 2.00	0.3%	negligible - and this is the region that works

Every target at or below 2.0 ATR is negative after costs, at every stop tested. Edge rises monotonically with both target and stop and the optimum sits **at the boundary of the grid** even after extending targets to 12 ATR. A boundary optimum that keeps improving is the mathematical statement of **no target at all** — which is precisely the shipped exit. The targets were never too large; they are always too small.

3. Optimisation and the three-way split

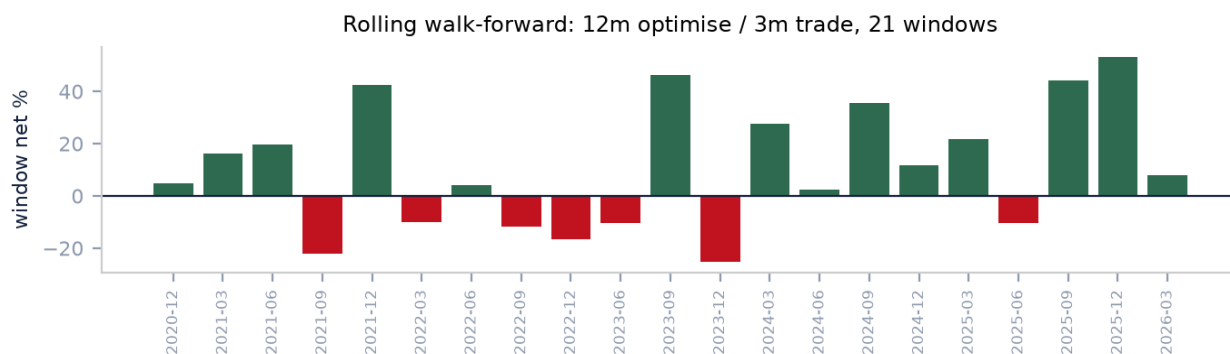
Item	Value
Combinations actually executed	4,500 (row count of opt_stage1_train.csv, not an estimate)
Passed the rejection filters	0
Rejected: profit factor < 1.15	1,839
Rejected: one month > 40% of profit	1,507
Rejected: drawdown > 55%	191
Train / Validation / Test	2019-12 to 2023-11 / 2023-11 to 2025-03 / 2025-03 to 2026-07

Stage	FROZEN (search-selected)	SHIPPED (untouched original)
Train (in-sample, 60%)	PF 1.326 / +561.5%	PF 1.324 / +507.0%
Validation (20%)	PF 1.737 / +111.6%	PF 1.484 / +75.0%
TEST (untouched, 20%)	PF 2.209 / +269.8% / DD 21.1%	PF 2.016 / +249.0% / DD 20.7%

Parameters and the selection rule were frozen to disk before the test window was read. **Trail 4.24 appears in all top 15 train configurations and entry 24 in 13 of 15** — the search re-finds the existing settings. Its total gain over changing nothing is **+0.193 profit factor**, which after 4,500 trials and 40 validated finalists is inside selection noise.

Both configurations improved from train to test rather than degrading. That is not vindication. The test window (Apr 2025 – Jul 2026) contains the largest gold trend in the sample. It is a favourable regime as much as it is evidence, and it should be read that way.

4. Walk-forward — re-optimising makes it worse



Approach	Compounded	Positive windows	Median window PF
Re-optimised every quarter (12m train / 3m trade)	+463.6%	14 / 21	1.430
Fixed shipped parameters, same 21 windows	+651.7%	14 / 21	1.314

Fixed parameters beat re-optimised ones in **11 of 21** windows. Twelve months is too short to estimate these parameters and the search picks up noise. This is evidence the parameters are **stable rather than fitted** — the opposite of what an overfit system shows.

5. Statistical edge

Measure	Value	Reading
Expectancy per trade	+1.451 R	1R = median losing trade
Bootstrap 95% CI	[+0.469, +2.592] R	5,000 resamples
P(expectancy <= 0)	0.08%	bootstrap, not a normal-theory p-value
Win rate / payoff	21.71% / 6.614	low hit rate is by design
Break-even win rate	13.13%	cushion of +8.58 points
Skew / kurtosis	+7.40 / +78.0	far from normal - trust the bootstrap
Trade autocorrelation	<= 0.063 at lags 1-5	trades are effectively independent

Risk of ruin and drawdown by risk per trade (5,000 bootstrap paths each)

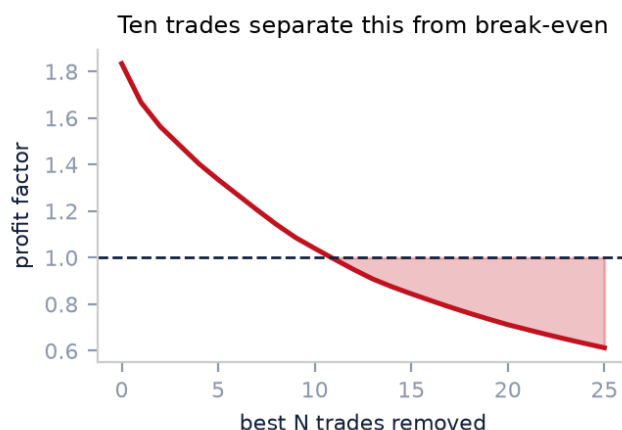
Risk / trade	Median DD	95th pct DD	P(DD over 50%)	P(ruin, -80%)	Median net
0.50%	19.4%	30.2%	0.0%	0.00%	+835%
0.75%	27.7%	42.2%	0.9%	0.00%	+2,309%
1.00% (shipped)	35.6%	53.2%	8.3%	0.00%	+5,941%
1.50%	49.0%	68.2%	46.0%	0.04%	+27,321%
2.00%	60.1%	78.9%	85.5%	0.24%	+91,014%

6. Cost sensitivity

Slippage per side	Full-sample PF	Full-sample net	Max DD	Test-window PF
0.00 pt	1.940	+9,950.9%	28.73%	2.065
0.20 pt (shipped)	1.834	+5,921.9%	33.80%	2.016
0.50 pt	1.691	+2,675.8%	41.82%	2.006
1.00 pt	1.364	+581.7%	66.67%	1.912
1.50 pt	1.087	+74.6%	85.84%	1.756

Survives a doubling of slippage to 0.40 per side (PF 1.716). Break-even is near 1.6 points. Execution quality remains the largest single risk and it is larger than any parameter in the system.

7. The finding that limits everything else



Best trades removed	Profit factor	Total P&L
0 (as traded)	1.834	+\$592,189
1	1.668	+\$474,198
5	1.335	+\$238,139
10	1.040	+\$28,230
20	0.712	-\$204,468
Best single MONTH (Mar 2026)	1.669	+\$461,471

Ten trades out of 737 separate this system from break-even. Twenty separate it from a losing one. Combined with a worst losing streak of 25 consecutive positions and an 8.3% chance of exceeding a 50% drawdown at 1% risk, this — not any parameter — is the constraint that governs whether the strategy is tradeable. It is inherent to trend following and no configuration tested removes it.

8. Regime, direction and session

Breakdown	Trades	Profit factor	Net P&L
Long	484	1.806	+\$384,477
Short	253	1.891	+\$207,712
Low volatility (ATR rank < 0.33)	300	1.632	+\$214,853
Mid volatility	217	1.709	+\$147,785
High volatility (rank > 0.67)	208	2.403	+\$224,563

The short side has the HIGHER profit factor despite a lower win rate. Cutting it was tested and rejected: it degrades exactly the two calendar years in which gold fell, and it is harmful on a second instrument.

Session (reported as a hypothesis, NOT applied)

Session (UTC)	Signals	Mean 50-bar return (ATR)	After 0.146 ATR cost
London 07-12	915	+1.111	+0.965
Asia 00-07	1,015	+0.926	+0.780
LDN/NY overlap 12-16	1,181	+0.838	+0.692
Late / rollover 21-24	205	+0.509	+0.363
New York 16-21	718	+0.041	-0.105 NEGATIVE

New York is the only losing session. **It has deliberately NOT been applied as a filter** because it has not been validated out of sample, and adding it now would be exactly the kind of in-sample fitting the rest of this study is designed to avoid. Timestamps are UTC; a fixed UTC hour drifts one hour against local session time twice a year and this has not been corrected.

9. Replication, assumptions and weaknesses

Item	Status
Repainting	None. No request.security, no varip, no realtime branches; all channel lookbacks [1]-offset.
Lookahead	None. Signals evaluate on closed bars; entry is the next bar's open.
Same-bar stop/target	Unresolvable at 30m. Resolved conservatively as losses; frequency reported per cell.
Pyramid fills	Gap-aware: if a bar opened beyond the add level, the fill is the open, not the level.
Data	No 5-minute gold data exists in this project. 15-minute data resampled to 30m was used, as directed.
Test-window regime	The untouched window contains gold's largest trend of the sample. Results are regime-flattered.
Sample size	737 positions over 6.7 years. Adequate for expectancy, thin for session and regime subgroups.
Overnight financing	NOT modelled. Average hold is 21 hours, so swap costs are real and would reduce these figures.

Exported files: signals_mfe_mae.csv (4,034 signals, full excursion detail), opt_stage1_train.csv (all 4,500 combinations), tp_sl_grid.csv and tp_sl_grid_extended.csv, stage1_finalists.csv, stage2_validation.csv, walkforward.csv, trades_shipped.csv (737 trades). Code in research/mfe_mae.py, tp_sl_grid.py, optimise.py, walkforward.py.

Generated 3 August 2026. All figures include 0.07 commission and 0.20 points of slippage per side unless stated. Backtested performance is not a prediction and this document is not proof of a future edge. Research record, not investment advice.